



Project 2: Beneficiary & Application Management

Team Name: Operations

Team Members: Hayk Matshkalyan and Lusine Karapetyan

Primary Responsibility: Core Business Logic, Data Entry, and Workflow Validation.

1. Domain Description

You are building the "Daily Driver" for Social Workers. This module handles the registration of vulnerable citizens and processes their applications for government aid against strict eligibility rules.

2. Data Model Requirements

- **Citizen:** Name, National ID (Unique), DOB, Address, *Annual Income (Sensitive)*.
- **AssistanceProgram:** Name, Active Status, Logic Rules (min_age, max_age, max_income_threshold).
- **Application:** Links a Citizen to a Program. Contains status (SUBMITTED, REVIEW, APPROVED, REJECTED) and submission_date.

3. Functional Requirements (MVP)

A. Citizen Registry

- **CRUD Operations:** Full management of Citizen profiles.
- **Data Privacy:** The Annual Income field must be masked (returned as *****) unless the user has the specific CITIZEN:VIEW_SENSITIVE permission in their JWT.
- **Search:** Implement searching by National ID or partial Name.

B. Program Catalog (Rule Engine)

- **Definition:** Allow managers to define programs (e.g., "Youth Support 2025").
- **Rules:** Managers set constraints here (e.g., Max Age = 25). These values drive the validation logic in the Application phase.

C. Application Workflow & Validation

- **Submission Logic:** When a Citizen applies for a Program, the backend must validate eligibility:
 - *Check:* Is Citizen.Age between Program.MinAge and Program.MaxAge?
 - *Check:* Is Citizen.Income < Program.IncomeThreshold?
 - *Result:* If checks fail, return 400 Bad Request with a specific error message.
- **Status Workflow:**
 - Social Workers can change status to REVIEW.
 - Only users with APPLICATION:APPROVE permission can transition status to APPROVED or REJECTED.

4. Optional & Advanced Features (Distinction)

-  **Internationalization (i18n):**
 - Frontend: Translate all form labels, table headers, and success/error messages.
 - Formatting: Display currencies and dates according to the selected locale (e.g., 10/12/2025 vs 2025-12-10).
- **Messaging/Audit:** When an application is REJECTED, publish a message to a mock Queue (RabbitMQ/Spring Events).
- **Family Logic:** Allow linking Citizens as "Parent/Child" and validate applications based on "Household Income" rather than individual income.
- **Drafts:** Implement a "Save as Draft" feature that bypasses validation rules until the user clicks "Final Submit".

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